

Food Delivery Subscription Customer Churn Analysis & Retention Strategy

Business Case Study | Customer Analytics | Predictive Modeling | Retention Strategy

Project Overview

The company's food delivery service had been losing many customers as of late due to recurring cancellations. This would affect their business by reducing recurring revenues and making it difficult for them to retain more clients.

The objective of this project is to analyze the behaviors, spendings, engagement, and satisfaction levels of their customers in order to determine what factors contribute to churn as well as create models that will be able to accurately predict which customers are likely to churn even before canceling subscriptions.

This project involves:

- Business Analysis
- Customer Behavior Analytics
- Predictive Modeling
- Customer Segmentation
- Retention Strategy Creation

Business Problem

Subscriptions require high levels of consistent revenue generation and customer engagement. Higher churn rates have significant impacts on profitability, CLV, and business expansion.

These were the main business questions that the organization sought to address:

- Which customers are most likely to cancel subscriptions?
- What are the key drivers of churn?
- Does customer satisfaction influence retention?
- Which customer segments are at highest risk?
- Can machine learning predict customer churn accurately?

Business Objective

The objective of this project was to:

- Analyze customer subscription behavior
- Determine factors leading to churning
- Predict high-risk customers using machine learning
- Segment customers based on churn probability
- Suggest customer retention strategies based on data
- Recommend data-driven retention strategies

Tools & Technologies

Tool	Purpose
Python	Data analysis & predictive modeling
Pandas & NumPy	Data preprocessing
Matplotlib & Seaborn	Data visualization
Scikit-learn	Machine learning models
Jupyter Notebook	Analytical workflow

Dataset Overview

- Food delivery subscription customer dataset
- 500+ customer records

Key Business KPIs

KPI	Value
Customer Churn Rate	21.8%
Customer Retention Rate	78.2%
Average Monthly Spend	₹814
Average Subscription Duration	8.56 Months
Random Forest Model Accuracy	93%

Business Analysis & Insights

1. Customer Spending Behavior Analysis

The analysis revealed that high spenders had low probabilities of churning.

Key Findings

Valued customers are more likely to stay loyal to the platform since they get better value out of it.

Impact on Business

Low-spend customers were at a very high risk of churning and hence contributed to volatile recurring revenue.

Recommendations

Implement:

- Loyalty programs
- Subscription perks
- Recommendations
- Incentives for engagement

These measures could help in increasing customer engagement with the platform.

2. Subscription Duration & Customer Loyalty

Subscribers with extended contract periods showed significantly higher retention than newer subscribers.

Key Findings

The greatest churn probability happened at the beginning of the customer journey.

Impact on Business

Low onboarding engagement led to high customer churn at early stages.

Recommendations

Enhance onboarding tactics by using:

- Welcoming deals
- Specials on free deliveries
- Customized onboarding campaigns
- Engagement promotions

3. Customer Satisfaction Analysis

The metrics that measure customer satisfaction such as app usage experience, delivery quality, and the money value ratio correlated strongly with customer churn behavior.

Key Findings

Satisfaction level measured by customers correlated strongly with likelihood of subscription cancellation.

Impact on Business

Quality service was directly linked to maintaining loyal customers and stabilizing revenue.

Recommendations

Prioritize areas that require improvement:

- Delivery reliability
- Rapid response from customer support services
- App performance optimization
- Value of the subscription

A good customer experience helps prevent churning.

4. Customer Engagement & Order Frequency

Frequent ordering of customers resulted in better retention tendencies.

Key Findings

Engaged platforms resulted in increased customer loyalty and lower chances of losing them.

Impact on Business

Low-engagement customers posed as major retention challenges.

Recommendations

Improvement in customer engagement by way of:

- Repeat-order rewards
- Loyalty programs with game mechanics
- Personalized offers
- Focused campaigns for engaging customers

5. Cancellation Intent Analysis

Users who expressed an intention to cancel their subscriptions were found to have a high probability of churn.

An intention to cancel was found to be a significant predictor of customer churn.

Impact on the Business

The company had no processes in place to intervene on behalf of at-risk customers.

Recommendation

Automated retention processes should be created to target users who express their intention to cancel their subscriptions through:

- Personal discounts
- Targeted Communication
- Customer support efforts
- Retention benefits

Predictive Analytics & Machine Learning

Objective

Develop predictive models capable of identifying customers at high risk of subscription cancellation.

Models Developed

Model	Accuracy
Logistic Regression	79%
Random Forest Classifier	93%

Model Performance & Business Value

The Random Forest model significantly outperformed Logistic Regression and successfully identified high-risk customers before subscription cancellation occurred.

Business Value of Predictive Modeling

The predictive model enables businesses to:

- Identify at-risk customers proactively
- Reduce recurring revenue loss
- Improve retention campaign effectiveness
- Prioritize high-value customer retention
- Optimize customer engagement strategies

Customer Risk Segmentation

Customers were categorized into three segments:

Risk Segment	Description
Low Risk	Loyal and highly engaged customers
Medium Risk	Moderate engagement with churn indicators
High Risk	Customers highly likely to cancel subscriptions

This segmentation enables targeted retention campaigns based on customer risk levels.

Key Churn Drivers Identified

The strongest drivers influencing churn were:

- Low customer satisfaction
- Weak value-for-money perception
- Low monthly spending
- Low order frequency
- Short subscription duration
- Cancellation intent

Recommendations

1. Implement Predictive Churn Monitoring

Implement a machine-learning-enabled churn monitoring system in order to detect the most likely to leave customers in advance. It will assist the organization in retaining its customers in an effective manner and reduce churn rates.

2. Optimize Customer Experience

It is recommended that the company concentrates on optimizing its delivery, improving the performance of its application, increasing the responsiveness of customer support, and offering additional benefits in terms of subscriptions. Such improvements may lead to the enhancement of customer satisfaction and loyalty which may decrease the probability of cancellation.

3. Develop Personalized Retention Strategies

The company should implement personalization in retention marketing efforts by offering customers customized offers, discounts, loyalty programs, and engaging them using special campaigns. It is possible that such an approach may lower the probability of churn and improve subscription-based revenue streams.

4. Enhance Early Customer Lifecycle Engagement

As new customers have exhibited the tendency to churn, it is critical that the company focuses on engaging its new customers during their first few months. Early engagement is likely to lower initial churn and enhance customer activation.

5. Establish Real-Time Customer Risk Dashboard

A real-time dashboard that monitors churn probabilities, customer engagement, satisfaction, and spending is highly recommended.

Future Enhancements

Potential future improvements include:

- Real-time churn prediction systems
- Customer lifetime value (CLV) modeling
- Power BI churn monitoring dashboard
- Automated retention campaign triggers
- Recommendation systems for personalized engagement
- Deep learning-based churn prediction

Conclusion

This project highlights the power of using customer analytics and machine learning to assist subscription-based companies in detecting possible churn risks and enhancing their retention capabilities.

Through behavioral analysis, predictive modeling, and business strategies, the project serves as a practical model for promoting customer loyalty, avoiding recurring revenue losses, and fostering company growth.